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June 21, 1999

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Division of Minerals & Geology

Ms. Christina Kamnikar
Colorado Division of Minerals and Geology
1313 Sherman Street, Room 215
Denver, Colorado 80203

Subject: Transmittal, "Report on Storage of Cement Kiln Dust,
Southdown, Inc. - Lyons Plant"
Permit No. M-77-208

Dear Christina:

Transmitted with this letter are three (3) copies of the above referenced report. Please let me know if you have any questions.

Sincerely,

Paul Banks

Paul Banks

PB/kj

CC: John Lohr, Southdown, Inc..

REPORT ON STORAGE OF CEMENT KILN DUST
SOUTHDOWN, INC. - LYONS PLANT

SOUTHDOWN, INC., LYONS QUARRY
PERMIT No. M-77-208

Prepared by:
Banks and Gesso, LLC
and
Grant Environmental, Inc.

Date: June 21, 1999

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1.0 Conclusions

As requested by the Colorado Division of Minerals and Geology, Southdown conducted an evaluation of their CKD management practices at the Lyons cement plant. This evaluation included a review of historical and current mining permits, characterization of cement kiln dust, assessment of local geology and hydrology, and an analysis of potential ground water impacts.

Conclusions from the above studies are summarized as follows:

1. Disposal of the CKD in C-Pit is allowed under CMLRB Permit No. M-77-208

2. Disposal of CKD in C-Pit presents little or no impact to ground water resources in the area.

3. Although the laboratory CKD samples showed several parameters which exceeded ground water standards, the two background samples (pasture soil and North 59th St.) also had parameters above the ground water standard.

4. There were five parameters from the laboratory CKD sample that exceeded applicable ground water standards by small levels. The parameters are:

    - selenium
    - thallium
    - total dissolved solids
    - pH
    - gross beta

    Geochemical computer modeling of the transport and fate of these five parameters over distance and time indicates compliance with all ground water standards before they could potentially migrate off of Southdown's property to the locally utilized aquifers.

5. The only useable ground water sources in this area are in the alluvial deposits of St. Vrain Creek and in the Dakota Sandstone. There is no hydraulic pathway from the C-Pit limestone and shale units to either of these ground water sources

6. The rock formations adjacent to and beneath C-Pit have relatively low permeability. The bedding planes in these sedimentary rocks do not provide a hydraulic connection with the alluvial or Dakota Sandstone aquifers. Permeability perpendicular to or across bedding planes is

extremely low. There is a lack of faulting in the area of C-Pit. These factors create a favorable CKD storage environment.

7. The testing, modeling and interpretation used in this research was very conservtive (i.e. worst case scenarios).

   - The model used the concentrated leachate from laboratory tests, not the much less concentrated water in the pit.
   - The model assumed the shale adjacent to C-Pit is saturated, which it is not
   - The model used conservative attenuation factors
   - The SPLP laboratory test is aggressive in that the CKD was ground up, mixed with a water solution having a lower pH than natural waters, and agitated for 24 hours.
   - The water flux rate (i.e. amount of water passing through the CKD in C-Pit) that was used in the model was conservative.

8. Southdown's management practice of periodically capping the disposed CKD with a layer of compacted shale or clay soils would further isolate the CKD from precipitation and infiltration of rain and snowmelt. Southdown has committed to doing this in cells that are built to 7 acres in size and in approximately 20 ft. lifts.

2.0 Introduction and Regulatory Background

Southdown, Inc. and its predecessors have operated a cement plant and associated quarries in Boulder County near Lyons since about 1969. Cement kiln dust (CKD), baghouse dust, and plant clean-up dust have been managed in the quarries since the inception of the cement plant. There are no known adverse impacts to groundwater from this historic practice.

The quarries south of the cement plant (referred to as the Lyons quarries) received a 112 permit for the Colorado Mined Land Reclamation Board in 1977. (Permit No. M-77-208). The permit is active and in good standing, the financial warranty is in place, and the facility is in compliance with the CMLRB permit. There are currently over 300 acres under active reclamation. Waste materials generated at a permitted mine site (the cement plant is within the permit area) may be disposed at the mine site without a Certificate of Designation pursuant to the requirements of the Colorado Department of Public Health and the Environment (CDPHE). Disposal of the CKD in the quarries is in compliance with the requirements of both the Division of Minerals and Geology (DMG) and the CDPHE.

On January 20, 1999 Carl Mount and Christina Kamnikar of the DMG inspected the facility. The inspection report requested a characterization of the chemical constituents of the CKD, a characterization of the local geology and hydrology, and an analysis of potential groundwater impacts. This report responds to that request.

Groundwater quality standards used in this analysis are either human health standards, secondary standards, agricultural standards or Federal drinking water standards, whichever is most restrictive. These ground water quality standards were selected based on discussions with Harry Posey of the DMG.

3.0 Generation and Disposal of Cement Kiln Dust

The Lyons cement plant currently generates approximately 20,000 - 30,000 tons of CKD annually. Cement kiln dust is made up of fine grained particulates of crushed and calcined limestone, shale, gypsum and silica. It contains the same constituents of the rocks that are in the vicinity of the plant. The cement kiln dust is removed from the kiln on a continuous basis when the kiln is operating. From there it is put into a pug mill and mixed with water prior to truck transport to the disposal area in the quarry. The addition of water keeps dusting to a minimum for the approximately 2000-foot haul distance into the quarry pit. The CKD is hauled in open dump trucks.

CKD is end dumped in C-pit. Permit number M-77-208 allows dust disposal in C-Pit. The disposed dust is frequently sprayed with water from a battery of large sprinklers surrounding the disposal area. The sprinklers place enough water on the dust to minimize wind erosion, but not enough to saturate the mass. There is some surface runoff of sprinkler water into the bottom of C-pit but the volume is relatively minor.

4.0 Chemical Characteristics of the Cement Kiln Dust

4.1 Sample Locations and Sampling Procedures

The chemical characteristics of the CKD were evaluated by sampling and laboratory analysis of the CKD presently disposed in C-Pit. Sampling and laboratory analysis was also performed on cement raw materials (limestone and 3rd Ridge Shale), kiln feed, a background soil sample from a local pasture upgradient of the plant, and roadbed material along North 59th Street downgradient of the plant. These samples were obtained and analyzed to provide a baseline to evaluate the results of the CKD analyses. Samples of water from A-Pit and a local water well (downgradient from the plant) completed in the Dakota Sandstone were also collected for comparison. A-Pit was chosen because it is downgradient for surface water drainage. The Dakota Sandstone is the primary domestic well aquifer in this area.

Sampling of the CKD and other solid and aqueous materials was performed by personnel from Banks and Gesso LLC and Grant Environmental, Inc. (Grant) on

Wednesday, April 7, 1999. The CKD and other solid materials were collected using a decontaminated shovel, placed into pre-cleaned five-gallon plastic buckets, and capped with plastic lids. The plastic buckets were labeled with the sample number, sample name, and date of collection.

In order to collect representative samples, a composite sample was prepared of each solid material by collecting the sample from several different locations at each sample site. The sampling shovel was decontaminated between samples to prevent cross-contamination of subsequent samples. The shovel was decontaminated by brushing with a steel brush to remove adhered solids, and washing the shovel with a brush and a solution of Alconox (a non-phosphate laboratory grade detergent). The shovel was then rinsed with deionized water and air dried prior to collecting the next sample.

A CKD sample (BG-1) was collected from the CKD previously disposed in C- Pit. The limestone (BG-2) and 3rd Ridge Shale (BG-3) samples were collected from the raw material stockpiles located south of the plant near C-Pit. The kiln feed sample (BG-4) was obtained from inside the cement plant. An upgradient background soil sample (BG-5) was collected from Loukenon's pasture located west of U. S. Highway 36. A sample of the roadbed material (BG-6) along North 59th Street was also collected. This sample location is situated about 8,500 feet downgradient (southeast) of C-Pit. Figure 1 shows the solid sample locations.

Water samples were also collected from A-Pit and the deep Dakota well (BG-1D). Figure 1 shows the water sample locations. These samples were collected and placed into pre-cleaned bottles provided by the analytical laboratory. Each bottle was labeled with the sample number, date and time, and sampler name at the sampling site. The samples were placed into an ice cooler with blue ice and maintained at approximately 4 degrees Centigrade (^{o}C) until delivery to the analytical laboratory. Both the solid and water samples were maintained under chain-of-custody until delivered to the analytical laboratory. The chain-of-custody procedures were followed to establish a written record of sample handling between the sample site and the laboratory.

The deep Dakota Well sample was collected from a well located about 6,000 feet southeast of C-Pit. The well is completed at a depth of about 1,130 feet in the Dakota sandstone. The well is artesian and is free flowing at the well head. The sample of the well water was collected from a plastic hose connected to the well head. Measurements of pH, temperature, and redox potential were performed at the wellhead until consistent, stable values were obtained. After the field measurements had stabilized the water sample was collected. An A-Pit water sample was collected along the western shore of the reservoir. Measurements of pH, temperature, and redox potential were also made at this sample location. A-Pit samples were collected by gentley submersing the sample bottles in the pit and capping the bottles while submersed.

4.2 Laboratory Analyses and Laboratory Quality Assurance/Quality Control (QA/QC)

4.2.1 Analyses Performed and Analytical Methods

The solid and water samples were hand-delivered on Thursday April 8, 1999 by Grant Environmental, Inc. to Barringer Laboratories Inc. (Barringer) in Golden, Colorado for analysis. The required analyses were specified in a letter transmitted with the samples from Grant to Barringer. The analyses performed included:

- A whole-rock analysis of the CKD (BG-1) to determine the total elemental composition of the dust. The CKD was analyzed for the constituents listed in Table 4.1. This table also includes the analytical method and detection limit.

- Leachate analysis of the CKD and other solid materials was performed using the U. S. Environmental Protection Agency (EPA) Synthetic Precipitation Leaching Procedure (SPLP), SW-846 Method 1312 to determine the concentration of constituents in the leachate. The resulting leachates were analyzed for the constituents listed in Table 4.2. This table also includes the analytical method and detection limit.

  The SPLP is a rotary agitation method where constituents in the sample material are leached with an amount of extraction fluid equal to 20 times the weight of the sample material. About 100 grams of sample material is used in each leaching test. Extraction fluid #2 was used for the extractions. Extraction fluid #2 consists of a 60/40 weight percent mixture of ~~sulfuric and nitric acids~~ added to deionized water until the pH is adjusted to 5.1±0.05. The sample material and extraction fluid are placed in a glass extraction vessel that is rotated at 30 ± 2 revolutions per minute for 18 ± 2 hours at a temperature of 23 ± 2 degrees Centigrade. After agitation, the extract liquid is decanted, filtered, and analyzed. The SPLP leaching solution is representative of the composition of precipitation for the western U. S. The leachates were simulated in the laboratory as they might form under natural conditions by leaching the solid sample with a synthetic precipitation solution.

- Analysis of the groundwater samples for the constituents listed in Table 4.2. This table also includes the analytical method and detection limit.

4.2.2 Laboratory Quality Assurance/Quality Control

With the exception of analyses for which no EPA analytical protocol was available, all analyses were conducted using EPA-approved methods specified in SW-846. Routine laboratory QA/QC procedures were used for the analyses. The QC measures included duplicate samples, matrix spike samples, and certified standards. Duplicate samples were analyzed to determine the reproducibility of the results. Matrix spike samples were analyzed to determine sample matrix characteristics. Spike analyses consist of adding a known amount of each analyte to a sample and determining the percent recovery. The concentration of each analyte relative to the known added concentration provides an estimate of the reliability of the analysis and the properties of the sample matrix. Laboratory certified standards were also used to verify the accuracy of the analytical results.

4.3 Analytical Results

4.3.1 Composition of the CKD

Table 4.3 summarizes the results of the whole rock CKD analysis. The laboratory results data package is included in Appendix A. The results of the analysis indicate that the CKD is primarily composed of calcium oxides and carbonates (expressed as CO_2 loss on ignition), silicon dioxide, lesser amounts of aluminum, sodium, and potassium oxides, and other trace metals. This composition is reflective of the composition of the natural rock materials (limestone, shale, and silica) that are used to make cement.

The CKD also contains small quantities of natural radionuclides that are the source of gross alpha and gross beta activities. The activities determined for the radionuclides are within the range of activities usually found in native rocks and soils in the area. The results of a CKD analysis conducted by the EPA in 1992 indicates that the predominant natural radionuclide in the CKD is potassium-40 (K-40). Less quantities of other natural radionuclides also occur. These include bismuth-214 (Bi-214), lead-214 (Pb-214), radium-226 (Ra-226), radium-228 (Ra-228), thorium-230 (Th-230), uranium-234 (U-234), and uranium-238 (U-238). K-40, Ra-228, and Bi-214 are the most likely sources of the gross beta activity. The other radionuclides are primarily alpha-emitters which is the source of the gross alpha activity.

4.3.2 Composition of the Leachate

The concentration of leachable constituents determined for the CKD and other solid materials are summarized in Table 4.4. The laboratory results data package is included in Appendix A. Colorado primary and secondary drinking water standards and agricultural water standards are included in the table for comparison. The most stringent water quality standards were selected for comparison. The constituent concentration values shown in bold on Table 4.4 exceed the applicable water quality standard. Not bold, but OK

The results of the CKD leachate analyses indicate that the primary leachable constituents are calcium, sodium, potassium, strontium, barium, sulfate, chloride, and fluoride. Trace quantities of chromium, lithium, molybdenum, silicon, zinc, arsenic, selenium, thallium, and phosphate were also detected. The CKD also contains leachable trace natural radionuclides (discussed above in Section 4.3.1) that are the source of the gross alpha and beta activities of the leachate.

Most of the constituents detected in the leachate are less than the respective water quality standard. Of the constituents analyzed, only five constituents exceeded the applicable water quality standard. **Selenium** (1.2), **thallium** (7.0), and **gross beta** (1.2) exceeded the primary human health groundwater standards. **Selenium** (3.1) also exceeded the agricultural water quality standard (0.02 milligrams per liter [mg/L]) which is more stringent than the human health standard (0.05 mg/L). Total dissolved solids (**TDS**; 5.1) and **pH** (1.5) exceeded the secondary human health groundwater quality standards. The source of the high TDS concentration is primarily due to the high concentration of constituents that do not exceed any of the water quality standards, such as calcium, sodium, potassium, strontium, barium, sulfate, chloride, and fluoride. The number in parentheses after each constituent name represents the ratio of the leachate constituent concentration to the water quality standard. These ratios indicate that these constituents, excluding selenium and thallium, are generally less than 2 times the applicable water quality standard.

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Table 4.1
Analyte List for the CKD Whole-Rock Analysis

Analyte	Analytical Method
Major Oxides	
SiO_2 (%)	LMF-ICP[1]
Al_2O_3 (%)	LMF-ICP
Fe_2O_3 (%)	LMF-ICP
MgO (%)	LMF-ICP
CaO (%)	LMF-ICP
Na_2O (%)	LMF-ICP
K_2O (%)	LMF-ICP
P_2O_5 (%)	LMF-ICP
TiO_2 (%)	LMF-ICP
MnO (%)	LMF-ICP
BaO (%)	LMF-ICP
Loss on Ignition [CO_2 + H_2O] (%)	LMF-ICP
Trace Constituents	
Antimony	AA[2]
Arsenic	AA
Boron	ICP[3]
Barium	ICP
Bismuth	ICP
Cadmium	ICP
Cobalt	ICP
Chromium	ICP
Copper	ICP
Fluoride	ICP
Lead	ICP
Lanthanum	ICP
Mercury (ug/kg)	HYG[4]
Molybdenum	ICP
Nickel	ICP
Selenium	ICP
Silver	ICP
Strontium	ICP
Thsllium	AA
Tungsten	ICP
Vanadium	ICP
Zinc	ICP
Total Sulfur (%)	Leco-BLI[5]
Total Sulfide (%)	Leco-BLI
Total Sulfate (%)	Leco-BLI
Total Carbon (%)	Leco-BLI
Total Organic Carbon (%)	Leco-BLI
Total Carbon Dioxide (%)	Leco-BLI
Nitrate as N	EPA 300.0
Gross Alpha (pCi/g)	EPA 900.0
Gross Beta (pCi/g)	EPA 900.0

[1] Analyzed using Lithium metaborate fusion (LMF) with inductively coupled plasma (ICP) procedure.

[2] Analyzed using atomic absorption method.

[3] Analyzed using inductively coupled plasma (ICP) procedure.

[4] Analyzed using hydride generation procedure.

[5] Analyzed using Leco-Barringer Laboratories Inc. (BLI) procedure.

Table 4.2
Analyte List for the SPLP Leachate and Groundwater Analyses

Analyte	Analytical Method	Method Detection Limit
All units are mg/L, unless otherwise noted.		
Aluminum	EPA 6010B	0.05
Barium	EPA 6010B	0.02
Beryllium	EPA 6010B	0.004
Boron	EPA 6010B	0.1
Calcium	EPA 6010B	0.2
Cadmium	EPA 6010B	0.005
Chromium	EPA 6010B	0.01
Cobalt	EPA 6010B	0.01
Copper	EPA 6010B	0.01
Iron	EPA 6010B	0.1
Lead	EPA 6010B	0.05
Lithium	EPA 6010B	0.02
Magnesium	EPA 6010B	0.1
Manganese	EPA 6010B	0.005
Molybdenum	EPA 6010B	0.01
Nickel	EPA 6010B	0.04
Potassium	EPA 6010B	5
Silicon	EPA 6010B	0.5
Silver	EPA 6010B	0.01
Sodium	EPA 6010B	1
Strontium	EPA 6010B	0.005
Titanium	EPA 6010B	0.01
Vanadium	EPA 6010B	0.01
Zinc	EPA 6010B	0.02
Arsenic	EPA 7060	0.003
Antimony	EPA 7041	0.005
Selenium	EPA 7740	0.005
Thallium	EPA 7841	0.002
Mercury	EPA 7470	0.0002
Fluoride	EPA 300.0	0.1
Chloride	EPA 300.0	1
Nitrate as N	EPA 300.0	0.1
Nitrite as N	EPA 300.0	0.1
Sulfate	EPA 300.0	5.0
Phosphate	EPA 365.1	0.06
Sulfite	EPA 377.1	1
Total Organic Carbon	EPA 415.1	2
Total Inorganic Carbon	EPA 415.1	1
Total Dissolved Solids	EPA 160.1	10
Specific Conductivity (*µmh/cm*)	EPA 120.1	1
pH (*standard units*)	EPA 150.1	0.01
Sulfide	EPA 376.1	1
Alkalinity as Bicarbonate	EPA 310.1M	1
Alkalinity as Carbonate	EPA 310.1M	1
Gross Alpha (*pCi/L*)	EPA 900.0	Variable
Gross Beta (*pCi/L*)	EPA 900.0	Variable

Table 4.3
Results of the CKD Whole Rock Analysis

Analyte	CKD (BG-1)
All units are mg/kg, unless otherwise noted.	
Major Oxides	
SiO_2 (%)	17.19
Al_2O_3 (%)	4.34
Fe_2O_3 (%)	2.13
MgO (%)	0.70
CaO (%)	43.49
Na_2O (%)	0.43
K_2O (%)	1.44
P_2O_5 (%)	0.16
TiO_2 (%)	0.18
MnO (%)	0.09
BaO (%)	0.39
Loss on Ignition (%)	29.6
Trace Constituents	
Antimony	< 2
Arsenic	10
Boron	7
Barium	218
Bismuth	< 1
Cadmium	15.3
Cobalt	6
Chromium	11
Copper	25
Fluoride	1035
Lead	28
Lanthanum	7
Mercury (ug/kg)	33
Molybdenum	12
Nickel	22
Selenium	20
Silver	1.8
Strontium	< 2
Thsllium	6
Tungsten	< 1
Vanadium	117
Zinc	84
Total Sulfur (%)	1.34
Total Sulfide (%)	0.092
Total Sulfate (%)	3.744
Total Carbon (%)	5.8
Total Organic Carbon (%)	0.796
Total Carbon Dioxide (%)	18.365
Nitrate as N	3.0
Gross Alpha (pCi/g)	47±26
Gross Beta (pCi/g)	32±18

$$\frac{1.86^{-5} \text{ mrem}}{\text{pCi}} = \frac{.0000186 \text{ mrem}}{\text{pCi}}$$

$$\frac{.0000186 \text{ mrem}}{1 \text{ pCi}} \quad \frac{x}{343 \text{ pCi}} = .00637$$

Table 4.4
Results of the SPLP Leachate Analyses

Parameter	Groundwater Standard	Std Type[6]	~~CKD BG-1~~	Limestone BG-2	3[rd] Ridge Shale BG-3	Kiln Feed BG-4	Pasture Soil BG-5	N 59 th Street BG-6	
	All reporting units are mg/L, unless otherwise noted.								
Aluminum	5	A	< 0.05	0.06	0.09	1.33	0.70	10.0	Al
Barium	2.0	H	1.49	0.35	0.1	0.15	0.33	1.66	
Beryllium	0.004	H	< 0.004	< 0.004	< 0.004	< 0.004	< 0.004	< 0.004	
Boron	0.75	A	< 0.1	< 0.1	< 0.1	0.1	0.2	0.8	B
Calcium	None	N	666	7.2	28.7	70.3	5.2	14.2	
Cadmium	0.005	H	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	
Chromium	0.1	H	0.04	< 0.01	< 0.01	0.01	< 0.01	< 0.01	
Cobalt	0.05	A	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	
Copper	0.2	A	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.02	
Iron	0.3	S	< 0.1	< 0.1	< 0.1	< 0.1	0.5	6.0	Fe
Lead	0.05	H	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	
Lithium	2.5	A	0.22	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	
Magnesium	None	N	< 0.1	0.9	2	< 0.1	0.5	2.6	
Manganese	0.05	S	< 0.005	< 0.005	< 0.005	< 0.005	0.008	0.043	
Molybdenum	None	N	0.11	< 0.01	0.02	0.04	< 0.01	< 0.01	
Nickel	0.1	H	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	
Potassium	None	N	< 5	< 5	< 5	5	< 5	6	
Silicon	None	N	0.5	1.4	1	9.6	4.6	24.4	
Silver	0.05	H	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	
Sodium	None	N	100	5	2	5	4	21	
Strontium	None	N	20.0	0.218	0.494	1.03	0.029	0.074	
Titanium	None	N	< 0.01	< 0.01	< 0.01	< 0.01	0.01	0.12	
Vanadium	0.1	A	< 0.01	< 0.01	0.02	0.07	0.01	0.02	
Zinc	2.0	A	0.06	< 0.02	< 0.02	< 0.02	< 0.02	0.32	Zn
Arsenic	0.05	H	0.005	< 0.003	< 0.003	< 0.003	< 0.003	0.006	
Antimony	0.006	H	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	< 0.005	
Selenium	0.02	A	**0.061**	< 0.005	0.008	0.019	< 0.005	< 0.005	
Thallium	0.002	H	**0.014**	< 0.002	< 0.002	< 0.002	< 0.002	< 0.002	
Mercury	0.002	H	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	
Fluoride	2.0	A	1.3	0.3	0.3	0.4	0.1	0.6	
Chloride	250	S	27	< 1	< 1	1	< 1	1	
Nitrate as N	10.0	H	< 0.5	0.7	0.02	0.03	0.2	0.7	
Nitrite as N	1.0	H	< 0.5	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	
Sulfate	250	S	53	7	70	87	2	10	
Phosphate	None	N	0.08	< 0.06	< 0.06	< 0.06	0.39	0.25	
Sulfite	None	N	< 1	< 1	< 1	2	< 1	< 1	
Total Organic Carbon	None	N	1	< 1	< 1	2	< 1	7	
Total Inorganic Carbon	None	N	< 1	2	2	< 1	2	< 1	
Total Dissolved Solids	400	S	**2050**	14	98	216	16	126	
Specific Conductivity (µmh/cm)	None	N	8940	80	197	471	46	128	
pH (standard units)	6.5 – 8.5	S	**12.38**	7.71	7.6	**10.95**	7.59	7.74	
Sulfide	None	N	< 2	< 2	< 2	< 2	< 2	< 2	
Total Alkalinity as CaCO3	None	N	2460	22	16	108	19	46	
Alkalinity as Bicarbonate	None	N	< 1	22	16	< 1	19	46	
Alkalinity as Carbonate	None	N	120	< 1	< 1	36	< 1	< 1	
Gross Alpha[7] (pCi/L)	15 pCi/L	H	~~26±28~~	0.0±1.1	0.0±1.5	0.0±2.8	0.0±1.1	1.6±1.7	
Gross Beta[8] (pCi/L)	4 mrem/yr	F	~~[illegible]~~	0.7±2.5	1.0±2.3	2.5±2.4	2.0±2.2	13±3	

0.9 ± 5.5 pCi/L 3.2 ± 5.8 pCi/L

[6] Std Type refers to the type of groundwater standard as follows: H = Human Health Standard; S = Secondary Standard; A = Agricultural Standard; F = Federal Drinking Water Standard; N = No Standard.

[7] Gross alpha is defined as gross alpha excluding contributions from radon and uranium.

[8] The primary human health groundwater standard for gross beta is a dose. The gross beta results are reported in pCi/L. The gross beta dose for BG-1, assuming that K-40 is the primary beta contributor, is 4.7 mrem/year. This value was calculated by assuming a groundwater ingestion rate of 2 liters per day for 365 days. The groundwater ingestion dose conversion factor was 1.86E-5 mrem/pCi.

Can't duplicate their gross Beta value

		[illegible]
		[illegible] Sh
U Cret	Niobrara	
		[illegible] LS
		[illegible] Sh
		Greenhorn LS
		Graneros Sh
		Mowry Sh
L. Cret	Dakota	

5.0 Geologic and Hydrogeologic Setting of the Disposal Area

The Lyons quarries are located entirely within the Upper Cretaceous Niobrara Formation. This Formation has a total thickness of about 400-feet and consists of two members, the Smokey Hill Shale Member and the Fort Hays Limestone Member. The Smokey Hill Shale is a fissile, calcareous shale and the Fort Hays Limestone is a thick-bedded limestone. In the quarry area the bedding planes dip to the east - southeast at about 10-15 degrees. C-Pit was excavated into the Fort Hays Limestone and the pit is underlain by the Carlile Shale, Greenhorn Limestone, Graneros Shale and Mowry Shale. These rock units consist of interlayered siltstone, claystones, limestones and shales. This sequence is about 500-feet thick and is underlain by the Lower Cretaceous Dakota Group. C-Pit is about 20 acres in size and 70-ft deep.

The area surrounding C-Pit has been heavily disturbed by mining. Mining and subsequent backfilling with overburden has occurred west and east of C-Pit, pursuant to the reclamation plan approved in M-77-208.

North and east of C-Pit are alluvial deposits of the St. Vrain Creek The sand and gravel deposits are 25-30 feet thick and come into contact with the sedimentary rocks surrounding C-pit approximately 1,000 feet away at the closest point. There is absolutely no contact between the disposed CKD and the alluvial deposit. The CKD disposal mass is entirely contained within the sedimentary rocks described above.

Significant surface water bodies in the area, and the approximate distance from the CKD disposal area in C-Pit are as follows:

St. Vrain Creek -	Approximately 4700' at the closest point
Foothills Reservoir -	Approximately 4200' at the closest point
A-Pit Reservoir -	Approximately 2000' at the closest point
(A-Pit Reservoir is an approved water storage facility on Southdown property and within the M-77-208 permit boundary. The water is used for the cement plant.)	
Swede Ditch -	Approximately 25' at the closest point
Boulder Feeder Canal -	Approximately 360' at the closet point

The Swede Ditch which runs along the west rim of C-Pit was all enclosed in a 4-foot diameter corrugated steel pipe in 1998 in preparation for conversion of C-Pit from active quarry to CKD storage. All the pipe joints were sealed. The Boulder Feeder Canal is approximately 360-feet west and 9.5-feet higher (elevation difference from the bottom of the canal compared to the rim of C-Pit) than the rim of C-Pit. This eliminates the possibility of gravity flow from C-Pit to the canal. Runoff from the disposed CKD can not escape C-Pit and is entirely contained within the pit. C-Pit CKD storage has been designed and situated to eliminate the potential for leachate seepage into agricultural water canals now and in the future.

Groundwater in the area occurs in both the sedimentary rocks and in the alluvial aquifer of the St. Vrain Creek. Groundwater levels in the sedimentary strata are well below the bottom of C-Pit. Ponded water currently in C-Pit is from surface runoff, collected precipitation, Boulder Feeder Canal seepage, and runoff from the sprinkler system. Southdown drilled two water wells immediately east of the permit area and had to advance the holes 800 feet or more to encounter a useable water supply. The exposed highwalls of C-Pit are dry except for seepage in response to rainfall or snowmelt, or from ditch seepage. The direction of groundwater flow in the sedimentary strata is to the east, primarily along bedding planes.

The western highwall of C-Pit was dry this year, before the water began to flow in the Boulder Feeder Canal. About 3 days after the Canal started flowing, seepage appeared in the C-Pit highwall. This fact indicates that groundwater is not the source of these observed seeps. Figures 3A, 3B and 3C show a sequence of photographs of the western highwall of C-Pit before and after the Canal started flowing.

There is no apparent hydraulic connection between C-Pit and the alluvial aquifer. The alluvial aquifer of St. Vrain Creek is in contact with the sedimentary strata east of C-Pit but not in contact with C-Pit itself. Groundwater in the alluvial aquifer is typically 5-8 feet below ground surface and flows east-southeast, generally towards St. Vrain Creek.

Further evidence of the lack of potential for leachate generation from the disposed CKD is provided by an analysis of the pH of the dust control water and precipitation runoff that collects along the C-Pit highwall downgradient of the disposed CKD. The water that collects along the highwall is either in direct contract with CKD, or is dust control water or precipitation that fell on the CKD and drained to this low area. If the CKD were not hydrated in its disposed state, the runoff water could be expected to have a much higher pH than natural waters that may occur in the area.

In order to determine the potential impact of the disposed CKD on area water quality, field pH measurements were made on April 7, 1999 at the edge of the water that had collected along the C-Pit hghwall. The pH measurements were taken in the water at approximate 100-foot intervals from the northeast to the southwest end of C-Pit. The water depth varied between about 6 inches to one foot at the measuring stations. The pH measurements ranged between 8.34 and 8.50 standard pH units. These pH values are similar to the 8.04 field pH measured at A-Pit. The results of the C-Pit pH survey suggest that the quality of water that contacts and runs off the disposed CKD is similar to the natural water quality in the area and that the disposed CKD does not pose a significant threat to groundwater quality.

6.0 Surrounding Wells and Local Aquifers

There are two aquifers in the area that are typically used for water supply. They are the alluvial aquifer of St. Vrain Creek and the Dakota sandstone. Both of these aquifers are isolated from C-Pit by 800-1000 feet of relatively impermeable sedimentary rocks.

Figure 4 shows a plot of wells in the area based on data from the Colorado Division of Water Resources (permitted wells database). The list of wells from the Colorado Division of Water Resources is included in Appendix B. Wells are shown by quarter - quarter section. It should be noted that many of the wells are piezometers and monitoring wells adjacent to Southdown's property in the sand and gravel alluvium.

Recent water quality data is available from wells in both the Dakota sandstone and in the alluvial aquifer. Water quality in both aquifers is generally good with no indication of any impact from CKD disposal.

A well completed in the Dakota sandstone (located in the extreme southeast corner of section 28) was sampled in April, 1999 by personnel from Grant Environmental, Inc. The laboratory results (Table 6.1) indicate that the water quality is in compliance with almost all applicable standards. Results of water quality analysis in A-Pit Reservoir are also shown in Table 6.1.

The pH (which would serve as a key indicator of impacts due to CKD disposal) of Dakota Well water is 8.28, within the natural, expected range. Table 6.1 shows the measured water quality in this well compared to the groundwater standards suggested by the DMG.

Eight wells completed in the alluvial aquifer have been recently sampled and analyzed for a variety of water quality parameters. These results are presented in Appendix C and they indicate good water quality that meets the various groundwater quality standards. There is no indication of elevated pH, which is evidence that the water in the alluvial aquifer is not affected by CKD disposal.

Figure 1 shows the location of the Dakota well and the eight alluvial wells.

7.0 Fate and Transport of Certain Constituents of Concern in Groundwater

7.1 Introduction

The hypothetical groundwater quality impacts resulting from the disposal of CKD in C-Pit are the formation and migration of leachate from the disposed CKD to groundwater. Leachate forms as precipitation infiltration percolates through the backfilled CKD and dissolves soluble constituents present in the CKD. The amount and composition of leachate is governed by three considerations: 1) the

Table 6-1
Groundwater Analyses Results

Parameter	Groundwater Standard	Std Type[1]	Deep Dakota Well (BG-1D)	A-Pit
	All units in mg/L, unless otherwise noted.			
Aluminum	5	A	0.07	0.05
Barium	2.0	H	0.03	0.03
Beryllium	0.004	H	< 0.004	< 0.004
Boron	0.75	A	0.3	0.2
Calcium	None	N	2.1	320
Cadmium	0.005	H	< 0.005	< 0.005
Chromium	0.1	H	< 0.01	< 0.01
Cobalt	0.05	A	< 0.01	< 0.01
Copper	0.2	A	0.05	< 0.01
Iron	0.3	S	3.6	0.1
Lead	0.05	H	< 0.05	< 0.05
Lithium	2.5	A	0.15	0.22
Magnesium	None	N	0.5	80.2
Manganese	0.05	S	0.025	0.046
Molybdenum	None	N	< 0.01	0.11
Nickel	0.1	H	< 0.04	0.12
Potassium	None	N	< 5	< 5
Silicon	None	N	8.6	1.5
Silver	0.05	H	< 0.01	< 0.01
Sodium	None	N	213	249
Strontium	None	N	0.096	6.51
Titanium	None	N	< 0.01	< 0.01
Vanadium	0.1	A	< 0.01	< 0.01
Zinc	2.0	A	0.08	0.14
Arsenic	0.05	H	< 0.003	0.008
Antimony	None	N	< 0.005	0.015
Selenium	0.02	A	< 0.005	0.007
Thallium	0.002	H	< 0.002	< 0.002
Mercury	0.002	H	< 0.0002	< 0.0002
Fluoride	2.0	A	2.6	0.5
Chloride	250	S	7	5
Nitrate as N	10.0	H	< 0.01	0.4
Nitrite as N	1.0	H	< 0.01	< 0.1
Sulfate	250	S	6	**1480**
Phosphate	None	N	0.08	< 0.06
Sulfite	None	N	1	1
Sulfide	None	N	5	3
Total Organic Carbon	None	N	< 1	3
Total Dissolved Solids	1.25 x Bkg	S	465	**2230**
Specific Conductivity (µmh/cm)	None	N	836	2500
pH (standard units)	6.5 – 8.5	S	8.28	8.03
Total Inorganic Carbon	None	N	99	31
Alkalinity as Bicarbonate	None	N	487	149
Alkalinity as Carbonate	None	N	25	< 1
Gross Alpha (pCi/L)	15	H	3.2±5.8	**80±22**
Gross Beta (pCi/L)	4 mrem/yr	F	0.9±5.5	**17±15**

[1] Std Type refers to the type of groundwater standard as follows: H = Human Health Standard; S = Secondary Standard; A = Agricultural Standard; F = Federal Drinking Water Standard; N = No Standard.

infiltration of water into and movement of water through the backfilled CKD, 2) the chemical composition of the CKD, and 3) the geochemical reactions between the CKD and the water.

To determine the potential impact that the backfilled CKD may pose to groundwater, a hypothetical leachate release to groundwater was simulated. Because the backfilled CKD is not expected to be saturated, the leachate formed will migrate predominantly downward and enter the groundwater system beneath C-Pit.

7.2 Simulation of the Potential Quantity of Leachate

The potential volume of leachate that may be formed at the CKD disposal site was evaluated using the Hydrologic Evaluation of Landfill Performance (HELP) model, version 3.07 (Schroeder, et.al., 1994). The leachate volumes predicted by theHELP model are used as the constituent release rates in the contaminant transport simulations described in Section 7.3.

The HELP (Version 3, developed by U.S. Army Corps of Engineers) model is a quasi-two dimensional deterministic water budget model of water movement across, into, through, and out of landfills. The model accepts climatologic, soil, and design data, and uses a solution technique that accounts for the effects of surface storage, runoff, infiltration, percolation, evapotranspiration, soil moisture storage, and lateral drainage. The program facilitates rapid estimation of the amounts of runoff, drainage, infiltration, and leachate that might be expected from a wide variety of landfill designs. The model is applicable to open, partially-closed, and fully-closed sites.

The CKD disposal site was simulated as a closed disposal site. The closed CKD disposal site is representative of the site following placement of the final cover material. The area of the disposal site modeled was 20 acres with a 75 foot highwall at the deepest portion of the C-Pit. Default and synthetic climatologic and default soil characteristics data were used to perform the HELP analyses. Climatologic data for Denver, Colorado were used in the analyses. The HELP modeling results are summarized in Table 7.1. HELP model output is included in Appendix D.

An initial HELP model run was made to determine the flux of leachate from the base of the disposed CKD. The model was designed to account for the approximately 600,000 cubic yards of CKD expected to be disposed at the site over the next 20 years. The final cover material was assumed to be 2 feet thick with a poor grass cover. The final slope of the cover material was assumed to be 2 percent. This disposal scenario is conservative as it minimizes the amount of evapotranspiration which allows more water to infiltrate the disposed CKD. The HELP model results indicate that the closed CKD disposal site will generate a leachate flux rate of approximately 0.74 inches per year at the base of the CKD. For a 20 acre disposal site, the annual leachate volume generated is about 53, 898 cubic feet.

An additional HELP model run was made to estimate the amount of leachate that might be transported vertically beneath the CKD disposal site to the top of the Dakota Sandstone. The underlying Niobrara Formation was simulated as a clay soil with low permeability (1 X 10-6 cm/s) to represent the interbedded shales present within the Niobrara. The shales are the primary controls on vertical groundwater flow beneath C-Pit. The results of this HELP model run indicate that the predicted leachate flux rate to the Dakota Sandstone is about 0.005 inches per year or an annual leachate volume of 366 cubic feet per year.

7.3 Simulation of Constituent Fate and Transport

7.3.1 Introduction

Although no viable groundwater pathway appears to exist between the CKD disposal site and adjacent aquifers, the potential impact of a hypothetical CKD leachate release on groundwater quality in the alluvium along St. Vrain Creek and the Dakota Sandstone was simulated. Constituent transport to the alluvium was assumed to occur through the weathered Niobrara Formation that occurs in the shallow subsurface. Groundwater flow was assumed to be to the east-northeast, towards St. Vrain Creek. The impact of a hypothetical leachate release on the alluvium was evaluated by assuming that a hypothetical recepotor well is completed in the alluvium along the eastern boundary of the plant about 3,000 feet east of C-Pit.

The impact of a hypothetical leachate release on the deep Dakota Sandstone aquifer beneath the disposal site was also evaluated. A hypothetical leachate release was assumed to occur at the base of the CKD and is transported vertically downward to the Dakota Sandstone. This model assumption is very conservative, as the rock formations lying between the base of the CKD and the top of the Dakota Sandstone consist of a large percentage of low permeability shales which would greatly impede vertical groundwater flow. The impact of the hypothetical leachate release on Dakota Sandstone groundwater quality was evaluated by assuming that a hypothetical receptor well is completed in the Dakota Sandstone along the eastern boundary of the plant, about 3,000 feet east of the C-Pit. Groundwater flow in the Dakota Sandstone was assumed to be towards the east.

Constituent transport in both aquifers was simulated using a quasi-analytical groundwater flow and solute transport model. Groundwater flow in the alluvium and Dakota Sandstone was assumed to occur as porous flow. Conservative aquifer parameters and assumptions were used to simulate the hypothetical CKD leachate release and subsequent constituent transport to the hypothetical receptor wells completed at the plant boundary. Solute transport simulations were performed for selenium, thallium, total dissolved solids (TDS), and gross beta (simulated as potassium-40). These constituents were selected because they were the only constituents present in the leachate that exceeded water quality standards. pH also exceeded the water quality standard and was simulated using a geochemical mixing model. The pH simulations are described in Section 7.3.5.

7.3.2 Description of the Model

The hypothetical leachate release to alluvium and Dakota Sandstone groundwater was simulated using MYGRT (Version 2.0). MYGRT was developed by Tetra Tech, Inc. (1989) to simulate the migration of inorganic or organic constituents in groundwater downgradient of a constituent source. The model is based on the advection-dispersion-retardation-decay equation. Processes simulated for inorganic constituents include advection, dispersion, and linear sorption (retardation). Both reactive and non-reactive constituents can be simulated by varying the retardation factor (Rd). Retardation is a measure of the partitioning of reactive solutes between the aqueous phase and the solid phase. A Rd of 1 simulates a non-reactive solute that is not attenuated and travels at the same rate as groundwater. Initial constituent concentrations can be modeled by specifying the solute concentrations at the downgradient edge of the source or as an areal source. If the constituent concentration is specified as an areal source, the model calculates the solute concentration in the aquifer by mixing the upgradient aquifer water with leachate based on its flux and composition. The areal source method was used in these analyses.

The assumptions inherent to the MYGRT (Tetra Tech, Inc. 1989) code include:

- MYGRT only simulates groundwater flow and solute transport in the saturated zone; unsaturated (vadose) zone transport is not considered;
- The groundwater seepage velocity remains constant over the distance of a given simulation;
- Dispersion is represented by Fick's Law. The coefficients remain constant for a given simulation;
- Sorption is treated as linear equilibrium partitioning between aqueous and solid phases: Sorption was not simulated in these analyses;
- Transport of a single solute species for a given simulation is adequate. Interactions between species are not considered.

7.3.3 Data Sources

The shortest distance to the property line was chosen as a conservative assumption. The geologic units between the C-Pit and the hypothetical receptor wells were assumed to be saturated and to exhibit uniform porous medium characteristics.

The aquifer parameters used in MYGRT are summarized in Table 7.2. Solute transport simulations to the alluvium were conducted using a hydraulic conductivity value of 2,000 feet per year for the weathered Niobrara Formation. This value was determined by packer tests conducted in the Niobrara Formation as part of an evaluation of the water storage potential of the A and C-Pits (Rocky Mountain Consultants, 1988). The hydraulic conductivity of the Dakota Sandstone was also assumed to be about 2,000 feet per year. A range of reasonable hydraulic gradients, 0.1, 0.05, and 0.01, were used in the simulations since the hydraulic gradient at the site is not known.. These values are typical of similar hydrogeologic systems. Longitudinal and transverse dispersion coefficients were estimated using an approach suggested by Gelhar and others (1985). The downgradient flow direction was assumed to be due east towards the St. Vrain Creek. The saturated thickness of the alluvium and weathered Niobrara was assumed to be about 30-feet. The saturated thickness of the Dakota Sandstone was assumed to be about 90-feet. The effective porosity for the weathered Niobrara and the Dakota Sandstone was assumed to be 0.25. This value is considered typical of weathered soils and sandstone units.

The hypothetical leachate release rate for the alluvium transport simulation was estimated to be about 0.07 feet per year. The leachate release rate was estimated using the HELP model described in Section 7.2. A leachate release rate of 0.03 feet per year was estimated for the Dakota Sandstone transport simulation. The leachate flux rate used in the Dakota Sandstone simulation is about 80 times higher than the leachate flux rate (0.0004 feet per year) calculated by the HELP model because it is the lowest value that can be simulated using MYGRT. The hypothetical receptor well constituent concentrations predicted in the Dakota simulations may therefore be overestimated by as much as 80 times. Leachate seepage was assumed to occur over the area of the C-Pit, which is estimated to be about 20 acres.

MYGRT simulations were conducted for selenium, thallium, TDS, and gross beta. These constituents were the only ones in the synthetic CKD leachate that exceeded the Colorado water quality standards. Gross beta was simulated as potassium-40. Potassium-40, a naturally occuring radionuclide in shales, was determined to be the predominant beta emitter in the CKD by an analysis conducted by the EPA in 1992. The concentrations of the leachate constituents used in the solute transport analyses are shown in Table 7.3. These concentrations were determined by laboratory analysis of the synthetic CKD leachate. Transport of these constituents was simulated by assuming that they are not attenuated. A Rd value of 1 was used in the analyses. An Rd of 1 simulates a non-reactive solute that is not attenuated and travels at the same rate as groundwater. Simulating the constituents as non-reactive provides the most conservative assumption for constituent transport.

7.3.4 Results of the MYGRT Simulations

The results of the solute transport simulations are summarized in Table 7.4. The results are shown as a relative increase in constituent concentration for the alluvium and the Dakota Sandstone. Table 7.4 shows the predicted concentrations for each constituent for the range of hydraulic gradients simulated. The solute transport analyses predict that a hypothetical leachate release will not impact groundwater quality, under the scenarios evaluated, in either the St. Vrain Creek alluvium or the Dakota Sandstone at a hypothetical receptor well located about 3,000 feet east of the C-Pit. The background concentration for these constituents in the alluvium and the Dakota Sandstone groundwater are shown on the table for comparison. The background values were obtained from water samples collected in the A-Pit and Dakota Sandstone (Table 6.1).

Solute attenuation or decay was not simulated in these analyses so that a conservative simulation was maintained. However, some of the constituents present in the CKD leachate may undergo attenuation or decay during transport. Metal cations, such as thallium or selenium, are reactive constituents that may be readily attenuated during transport if sorptive solid phases such as clay minerals or iron and manganese hydroxides are present in the aquifer matrix. Because of the low concentrations of metals anticipated in a potential leachate release, the presence of sorptive materials in the aquifer matrix would rapidly attenuate most cations released to groundwater.

7.3.5 Mixing Simulation of pH in the Alluvium and Dakota Sandstone

Evaluation of the impact of a hypothetical CKD leachate release on alluvium and Dakota Sandstone groundwater pH was performed using PHREEQC (Parkhurst, 1995).✓PHREEQC is a computer program that performs speciation, mixing, reaction-path, advective-transport, and inverse geochemical calculations. PHREEQC uses an ion-association model and Debye Huckel expressions to account for the non-ideality of aqueous solutions. The speciation and mixing modules of PHREEQC were used in this analysis. In the mixing module, each solution is multiplied by its mixing fraction and a new solution is calculated by summing over all of the fractional solutions.

The mixing ratio of the groundwaters and the CKD leachate were determined using the annual leachate volumes calculated by the HELP model and the volume of each aquifer immediately beneath the CKD disposal site, an area of approximately 20 acres. The volume of water in the alluvium and Dakota Sandstone aquifers is estimated at 28,575,360 cubic feet and 85,726,080 cubic feet, respectively. The annual CKD leachate volumes estimated using HELP for the alluvium and Dakota Sandstone simulations are 53,898 cubic feet and 366 cubic feet, respectively. The initial PHREEQC calculations were performed using a mixing ratio of one part aquifer water volume to one part of the annual CKD leachate volume.

The results of the groundwater-leachate mixing simulations indicate that at reasonably expected volumes of CKD leachate, the alluvium and Dakota Sandstone aquifer pH will not be adversely impacted by a hypothetical CKD leachate release. At higher CKD leachate volumes (which are not reasonably expected to occur but were used for worst case analysis) the alluvial water quality may be adversely impacted. Once plant operations are completed, the capping and closure of the CKD storage area will prevent the hypothetical, high leachate volumes from occurring.

7.3.6 Summary

The results of the solute transport analyses suggest that a hypothetical CKD leachate release will not adversely impact groundwater quality in either the St. Vrain Creek alluvium or the underlying Dakota Sandstone. The solute transport model predicts that the hypothetical leachate release will be sufficiently dispersed during transport so that the existing groundwater quality in the alluvium or the Dakota Sandstone is not adversely affected. The solute transport simulations were performed using conservative model parameters. Under these conservative conditions and assuming no constituent attenuation, the concentration of leachate constituents released to groundwater do not cause the most stringent Colorado primary or secondary drinking water standards or agricultural water quality standards to be exceeded at the hypothetical receptor wells completed in the alluvium or the Dakota Sandstone.

Table 7.1

Summary of the HELP Model Results

Scenario	Leachate Release Rate (in/yr)	Leachate Volume (cu ft/yr)
Base of Disposed CKD	0.742	53,898
Top of Dakota Sandstone	0.005	366

Table 7.2

MYGRT Model Parameters

Parameter	Weathered Niobrara	Dakota Sandstone
Hydraulic Conductivity (m/y)	610	610
Hydraulic Gradient	0.01, 0.05, 0.1	0.01, 0.05, 0.1
Effective Porosity	0.25	0.25
Groundwater Velocity (m/y)	24, 122, 244	24, 122, 244
Longitudinal Dispersion (m^2/y)	2233, 11163, 22326	2233, 11163, 22326
Transverse Dispersion (m^2/y)	223, 1116, 2233	223, 1116, 2233
Width of Source (m)	350	350
Mixed Depth of Aquifer (m)	10	30
Volumetric Water Content	0.25	0.25
Retardation Coefficient	1	1
Net Leachate Infiltration Rate (m/y)	0.02	0.01 (model minimum)

Table 7.3

CKD Leachate Constituent Concentrations Used in MYGRT

Parameter	Leachate Concentration	
Selenium (mg/L)	0.061	0.02
Thallium (mg/L)	0.014	0.002
Total Dissolved Solids (mg/L)	2050	1.25 x BKG
Gross Betá (pCi/L)	347	4.0 mrem/y

Table 7.4
Constituent Transport Modeling Results

Results for Hydraulic Gradient = 0.1

Parameter	Predicted Increase in Alluvium Groundwater Concentrations	Alluvium Groundwater Background	Predicted Increase in Dakota Groundwater Concentrations	Dakota Groundwater Background	Colorado Water Quality Standard
Selenium (mg/L)	0.0004	< 0.002	0.0002	< 0.005	0.02
Thallium (mg/L)	0.00009	NA	0.00004	<0.002	0.002
TDS (mg/L)	12.9	680	6.5	465	1.25 x Bkg
Gross Beta (mrem/y)	0.03	NA	0.01	ND	4.0

Results for Hydraulic Gradient = 0.05

Parameter	Predicted Increase in Alluvium Groundwater Concentrations	Alluvium Groundwater Background	Predicted Increase in Dakota Groundwater Concentrations	Dakota Groundwater Background	Colorado Water Quality Stanard
Selenium (mg/L)	0.0008	< 0.002	0.0004	< 0.005	0.02
Thallium (mg/L)	0.0002	NA	0.00009	<0.002	0.002
TDS (mg/L)	25.5	680	12.8	465	1.25 x Bkg
Gross Beta (mrem/y)	0.06	NA	0.03	ND	4.0

Results for Hydraulic Gradient = 0.01

Parameter	Predicted Increase in Alluvium Groundwater Concentrations	Alluvium Groundwater Background	Predicted Increase in Dakota Groundwater Concentrations	Dakota Groundwater Background	Colorado Water Quality Standard
Selenium (mg/L)	0.004	< 0.002	0.002	< 0.005	0.02
Thallium (mg/L)	0.0008	NA	0.0004	<0.002	0.002
TDS (mg/L)	120.1	680	62.3	465	1.25 x Bkg
Gross Beta (mrem/y)	0.28	NA	0.14	ND	4.0

Table 7.5

Geochemical Mixing Model Results

Groundwater Mixed with CKD Leachate	Mixture Ratio (Groundwater:Leachate)	Simulated pH of Groundwater
Alluvium – 1V CKD Leachate	1 : 2E-3	7.58
Alluvium – 10V CKD Leachate	1 : 2E-2	8.79
Alluvium – 100V CKD Leachate	1 : 2E-1	11.10
Dakota Sandstone – 1V CKD Leachate	1 : 4E-6	8.28
Dakota Sandstone – 10V CKD Leachate	1 : 4E-5	8.28
Dakota Sandstone – 100V CKD Leachate	1 : 4E-4	8.31

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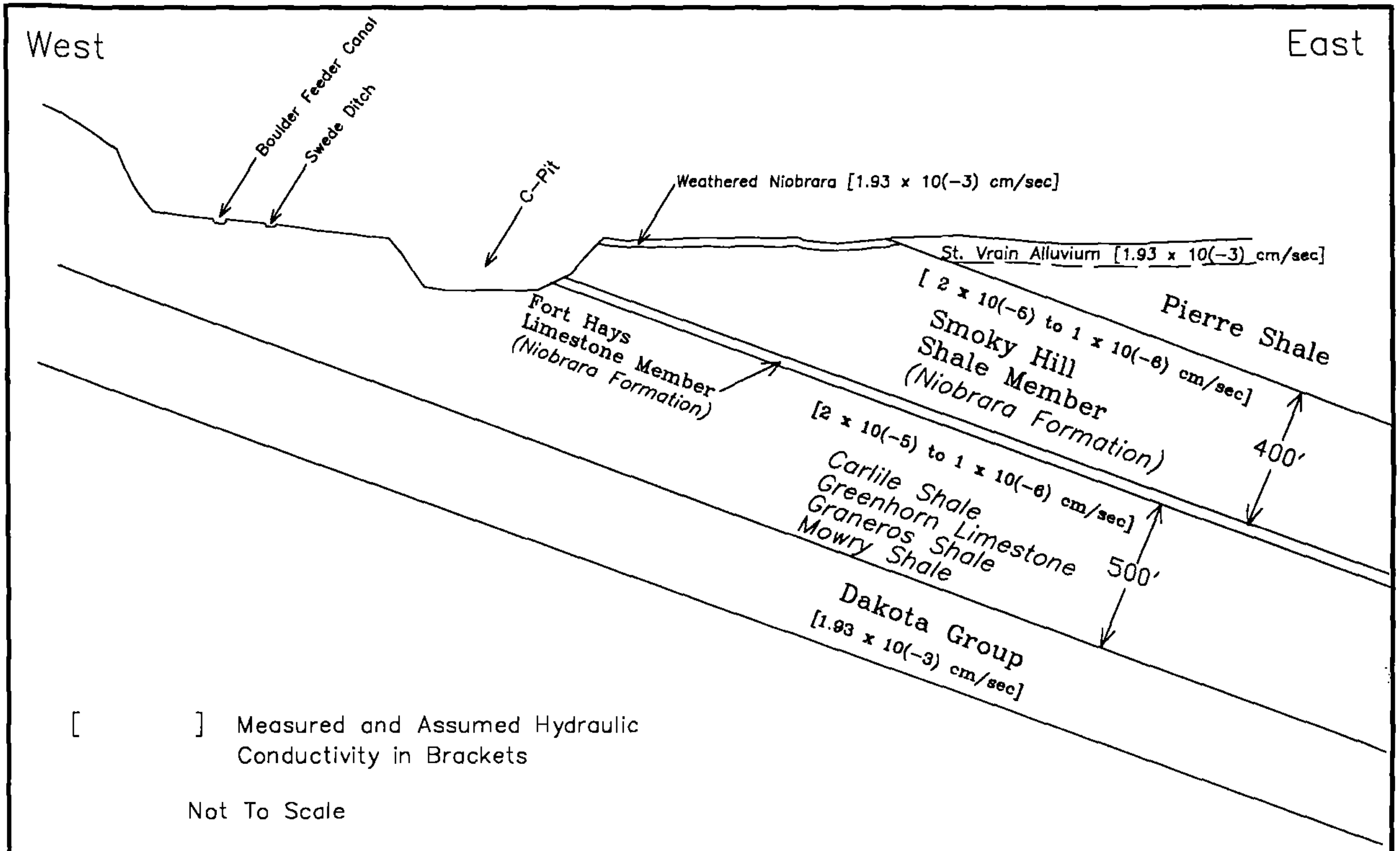

Banks & Gesso, LLC

Southdown, Inc.
Cement Kiln Dust Storage

Figure 2: Generalized Stratigraphic Cross-Section

Project #:	Date:	Scale:
99004	6/1/99	NA

APPENDIX A

Laboratory Results

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Project: Banks and Gesso PO #:

Job: 991821E Status: Final

ANALYTICAL REPORT PACKAGE

BARRINGER LABORATORIES, INC.
15000 W. 6TH AVE., SUITE 300 GOLDEN, CO 80401 (303) 277-1687 FAX (303) 277-1689

28-Apr-99

Mr. Paul Banks
Banks and Gesso LLC
720 Kipling Street
Suite 117
Lakewood, CO 80215

Attn:
Project: Banks and Gesso PO #:
Received: 8-Apr-99 10:45

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CASE NARRATIVE

AMENDED REPORT

A total of 8 Water/Solid samples were received on 8-Apr-99. As stated in the chain of custody, the samples were run for the following analyses: Al, Ba, Be, B, Ca, Cd, Cr, Co, Cu, Fe, Pb, Li, Mg, Mn, Mo, Ni, K, Si, Ag, Na, Sr, Ti, V, Zn, As, Sb, Se, Tl, Hg, F, Cl, NO3, NO2, SO4, PO4, SO3, Sulfide, TOC, TDS, Spec. Cond., pH, TICarbon, Alk-HCO3, Alk-CO3, Gross Alpha, Gross Beta and Gamma Spectroscopy. A table, to cross reference your sample ID to ours, is attached. Our procedures are summarized on the Quality Control Data Sheet.

Quality control standards for organic and inorganic analyses followed the appropriate SW-846 or EPA methodology. Quality control standards for radiochemistry followed our standard operating procedures or contractual requirements.

Some metals results have been determined by the method of standard additions (MSA) and are flagged with an asterisk.

The temperature range for the SPLP extraction was 20-25°C.

Sample 991821-3 was diluted for selenium due to matrix interferences. The MDL has been raised for this dilution.

Samples 991821-1 and 991821-2 required a dilution to accurately determine total inorganic carbon.

Sample 991821-2 required a dilution to accurately determine sulfate.

Sample 991821-3 was diluted for anions by method 300.0 due to matrix interferences. The MDLs have been raised for this dilution.

This report has been amended from the report dated April 22, 1999 to report a lower detection limit for thallium and to report the reanalysis of sample 991821-6 for conductivity.

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Signed: 4/29/98 4/29/99
Inorganic Manager   Radiochemical Manager

Signed:
Project Review 4/29/99